

curriculum vitae of
Filippo Mazzoli

UNIVERSITY OF CALIFORNIA RIVERSIDE, DEPARTMENT OF MATHEMATICS
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ANAGRAPHS

Born on November 21st, 1992 in Firenze (FI), Italy. Italian citizen. Pronouns: *he/him/his*.

EMPLOYMENT

Aug 2024 – present	Assistant Professor in Topology, Department of Mathematics, University of California Riverside, Riverside CA, USA. Tenure track position.	UC RIVERSIDE
Aug 2023 – Jul 2024	Postdoctoral researcher in Mathematics, Max Planck Institute for Mathematics in the Sciences, Leipzig, Germany. Post-doc mentor: Anna Wienhard.	MPI FOR MIS
Aug 2020 – May 2023	Whyburn Research Associate and Lecturer , Department of Mathematics, University of Virginia, Charlottesville VA, USA. Post-doc mentor: Sara Maloni.	UNIVERSITY OF VIRGINIA

EDUCATION

Oct 2016 – Jul 2020	Ph.D. in Mathematics, Thesis title: <i>Constant curvature surfaces and volumes of convex co-compact hyperbolic manifolds</i> , Advisor: Jean-Marc Schlenker. (pdf file available here .)	UNIVERSITÉ DU LUXEMBOURG
Sept 2014 – Sept 2016	M.Sc. in Mathematics, $\frac{110}{110}$, summa cum laude. Thesis advisor: Riccardo Benedetti.	UNIVERSITÀ DI PISA
Sep 2011 – Sep 2014	B.Sc. in Mathematics, $\frac{110}{110}$, summa cum laude. Thesis advisor: Paolo Lisca.	UNIVERSITÀ DI PISA

RESEARCH INTERESTS

(Higher) Teichmüller Theory, Anti-de Sitter geometry, Hyperbolic geometry, Kleinian groups,
Low-dimensional topology, quantum Teichmüller space, quantum hyperbolic invariants.

PAPERS

Published	1. S. Maloni, G. Martone, F. Mazzoli, and T. Zhang. <i>d-Pleated surfaces and their shear-bend coordinates</i>. Proc. London Math. Soc. 133.1, 2026, e70174. URL. 2. F. Mazzoli and G. Viaggi. <i>Volume, entropy, and diameter in $SO(p, q + 1)$-higher Teichmüller spaces</i>. Comment. Math. Helv. 101.3, 2026, pp. 555–596. URL. 3. F. Mazzoli, A. Seppi, and A. Tamburelli. <i>Para-hyperKähler geometry of the deformation space of maximal globally hyperbolic anti-de Sitter three-manifolds</i>. Mem. Amer. Math. Soc. 306.1546, 2025, pp. v+107. URL. 4. F. Mazzoli and G. Viaggi. <i>$SO_0(2, n + 1)$-maximal representations and hyperbolic surfaces</i>. Mem. Amer. Math. Soc. 313.1587, 2025, pp. v+119. URL. 5. F. Mazzoli. <i>Constant Gaussian curvature foliations and Schläfli formulas of hyperbolic 3-manifolds</i>. Ann. Sc. norm. super. Pisa - Cl. sci. 2024. URL. 6. F. Mazzoli and G. Viaggi. <i>Length functions in Teichmüller and anti-de Sitter geometry</i>. Forum Math. Sigma 11, 2023, Paper No. e101. URL.
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7. F. Mazzoli. *The infimum of the dual volume of convex co-compact hyperbolic 3-manifolds*. Geom. Topol. 27.6, 2023, pp. 2319–2346. [URL](#).
8. D. Choudhury, F. Mazzoli, and A. Seppi. *Quasi-Fuchsian manifolds close to the Fuchsian locus are foliated by constant mean curvature surfaces*. Math. Ann. 2023. [URL](#).
9. F. Mazzoli. *The dual volume of quasi-Fuchsian manifolds and the Weil-Petersson distance*. Trans. Amer. Math. Soc. 375.1, 2022, pp. 695–723. [URL](#).
10. F. Mazzoli. *The dual Bonahon-Schläfli formula*. Algebr. Geom. Topol. 21.1, 2021, pp. 279–315. [URL](#).
- Preprints 11. S. Maloni, G. Martone, F. Mazzoli, and T. Zhang. *Topology of the space of d -pleated surfaces*. submitted, 2025. arXiv: [2508.04813 \[math.GT\]](#).
12. F. Mazzoli. *Intertwining operators of the quantum Teichmüller space*. under revision, 2016. arXiv: [1610.06056 \[math.GT\]](#).

AWARDS AND GRANTS

2020	Excellent Thesis Award, year 2020.	UNIVERSITÉ DU LUXEMBOURG
2020	Graduate Internship grant.	GEAR NETWORK
2019	Travel grant for graduate students.	DPMA, UNIVERSITÉ DU LUXEMBOURG
2017	Premio di Studio Benedetto Sciarra, anni 2015-2016.	SCUOLA NORMALE SUPERIORE, PISA

LONG RESEARCH VISIT

Jan 2019 – Mar 2019	Supervised by Prof. Francis Bonahon.	UNIVERSITY OF SOUTHERN CALIFORNIA
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INVITED TALKS

CONFERENCES AND WORKSHOPS

May 23 rd , 2025	<i>Higher rank geometric structures, Higgs bundles and physics</i> , conference.	INSTITUT HENRI POINCARÉ
Apr 28 th -May 2 nd , 2025	<i>Leaning towards higher rank</i> , conference.	UNIVERSITÀ DI PISA
Apr 15 th -19 th , 2024	<i>Submanifolds of locally symmetric spaces: geometric, analytic and dynamical aspects</i> , workshop.	AUTRANS, FRANCE
Feb 21 st -23 rd , 2024	<i>Mini-workshop on pseudo-Riemannian geometry</i> , workshop.	UNIVERSITÀ DEGLI STUDI DI PAVIA
Jul 2023	<i>Group actions and low-dimensional topology</i> , conference.	ICMAT, EL BARCO DE ÁVILA, SPAIN
Jan 27 th , 2023	<i>Geometry, topology, and their applications</i> , online workshop.	MPI FOR MiS, LEIPZIG, GERMANY
Aug 22 nd -26 th , 2022	<i>Minimal surfaces in symmetric spaces and Labourie's conjecture</i> , workshop.	AUTRANS, FRANCE
Jul 20 th , 2022	<i>AMS-EMS-SMF Meeting</i> , conference.	UNIVERSITÉ GRENOBLE ALPES
Mar 12 th , 2022	<i>55th Spring Topology and Dynamics Conference</i> , conference.	BAYLOR UNIVERSITY
Jun 2021	<i>Nearly Carbon Neutral Geometric Topology Conference</i> , online conference.	ONLINE
Aug 2019	<i>Combinatorial and algebraic aspects of geometric structures</i> , conference.	CHIANG MAI UNIVERSITY
Dec 8 th , 2020	<i>Parole aux jeunes géomètres dynamiques</i> , online workshop.	GDR PLATON
Sep 2018	<i>Young Mathematicians Symposium of the Greater Region</i> , conference.	INRIA NANCY - GRAND EST
May 2018	<i>Structures on Surfaces</i> , workshop.	INRIA NANCY - GRAND EST

MINI-COURSES

Sept 2026	TBA, Research program.	LMSI INSTITUTE
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SEMINARS

June 16 th , 2026	Seminario di Geometria.	UNIVERSITÀ DEGLI STUDI DI TORINO
Mar 18 th , 2024	Discrete groups and geometry.	IHES
Mar 12 th , 2024	Geometry and Topology Seminar.	UNIVERSITÉ DU LUXEMBOURG
Feb 15 th , 2024	Séminaire de Théorie Spectrale et Géométrie.	INSTITUT FOURIER, GRENOBLE
Oct 10 th , 2023	Geometry Seminar.	MPI FOR MiS
Mar 6 th , 2023	Dynamics Seminar.	UNIVERSITY OF WISCONSIN-MADISON

Jan 13 th , 2023	Colloquium talk.	UC RIVERSIDE
Nov 28 th , 2022	Topology Seminar.	UT AUSTIN
Nov 18 th , 2022	Geometry-Topology Seminar.	UC RIVERSIDE
Apr 4 th , 2022	Geometry Seminar.	UNIVERSITY OF VIRGINIA
Nov 29 th , 2021	Geometry-Topology Seminar.	UNIVERSITY OF MARYLAND
Jun 10 th , 2021	Differential Geometry Seminar (online).	HEIDELBERG UNIVERSITY
May 11 th , 2021	Geometry Seminar.	UNIVERSITY OF VIRGINIA
Nov 3 rd , 2020	Pangolin Seminar.	ONLINE
Oct 14 th , 2020	Max Dehn Seminar.	UNIVERSITY OF UTAH
Feb 12 th , 2020	Topology Seminar.	TEXAS A&M
Jan 30 th , 2020	Geometry & Topology Seminar.	BOSTON COLLEGE
Jan 28 th , 2020	Geometry & Topology Seminar.	YALE UNIVERSITY
Jan 23 rd , 2020	Differential Geometry Seminar.	HEIDELBERG UNIVERSITY
Dec 12 th , 2019	Séminaire de Théorie Spectrale et Géométrie.	INSTITUT FOURIER, GRENoble
Feb 19 th , 2019	Geometry Seminar.	UNIVERSITY OF VIRGINIA

TEACHING

Winter 2026	MATH 011, Introduction to Discrete Structures, undergraduate course.	UC RIVERSIDE
	MATH 010A, Calculus of Several Variables, undergraduate course.	
Fall 2025	MATH 259, Geometry and topology of surfaces, graduate course.	
Winter 2025	MATH 010B, Calculus of Several Variables (part 2), undergraduate course.	
	MATH 009B, First-year Calculus (part 2), undergraduate course.	
Fall 2024	MATH 009B, First-year Calculus (part 2), undergraduate course.	
Jan 2024 – Feb 2024	An introduction to anti-de Sitter geometry, mini-course.	MPI FOR MfS
Spring 2023	MATH 8720, Hyperbolic Geometry, topics course.	UNIVERSITY OF VIRGINIA
	MATH 5700, Introduction to Geometry, undergraduate course.	
Fall 2022	MATH 3310-100, Basic Real Analysis, undergraduate course.	
Spring 2022	MATH 8720, Differential Geometry, graduate course.	
	MATH 5700, Introduction to Geometry, undergraduate course.	
Fall 2021	MATH 3310-100, Basic Real Analysis, undergraduate course.	
Spring 2021	MATH 3350-001, Applied Linear Algebra, undergraduate course.	
	MATH 3350-002, Applied Linear Algebra, undergraduate course.	
Fall 2020	MATH 2310-101/201, Calculus III (TA), undergraduate course.	
Fall 2019	BASI-313, Analyse 1 (TA), undergraduate course.	UNIVERSITÉ DU LUXEMBOURG
Fall 2018	BASI-280, Analyse 1a & 1b (TA), undergraduate course.	
Spring 2018	BASI-10 Analyse 2a & 2b (TA), undergraduate course.	
Fall 2017	BASI-12, Analyse 3b (TA), undergraduate course.	

MENTORING AND OUTREACH

Jan 2026-current	PhD Thesis, Mentee: David Weed. Topic: <i>TBD</i> .	UC RIVERSIDE
Jan-Jun 2024	ENS Lyon Internship, Mentee: Antoine Ablondi. Topic: <i>Pseudo-Riemannian hyperbolic geometry</i> .	MPI MfS
Oct 2022	Arbeitsgemeinschaft on Higher Teichmüller theory, Mentee: Fernando Camacho Cadena. Topic: <i>Symplectic structures on the Hitchin component</i> .	OBERWOLFACH
Oct 2022	Geometry Lab, Topic: <i>Hyperbolic geometry</i> . Organizer: Sara Maloni.	UNIVERSITY OF VIRGINIA

Summer 2021 Research Experience for Undergraduates (REU), UNIVERSITY OF VIRGINIA
 Co-advisor: Sara Maloni. Students: K. Betts, T. Larsen, J. Utley, A. Vanis.
 Project: *The tri-pants graph of the twice-punctured torus*. Preprint [arXiv:2111.07136](https://arxiv.org/abs/2111.07136).

2018-19 Science Comic: Through the Looking Glass, Reflections of Science in Luxembourg, UNIVERSITÉ DU LUXEMBOURG
 Title: *Number theory in the past, present, and future*. Comic book available [here](#).

ORGANIZING

SEMINARS

2025-2026 Geometry & Topology seminar, UC RIVERSIDE, RIVERSIDE CA, U.S.A.
 Co-organizers: Brian Collier.
 2025-2026 Colloquium talks committee, UC RIVERSIDE, RIVERSIDE CA, U.S.A.
 Co-organizers: Brian Collier, Tom Gannon, Siting Liu, Qi Zhang (chair).
 2024-2025 Colloquium talks committee, UC RIVERSIDE, RIVERSIDE CA, U.S.A.
 Co-organizers: Yat Tin Chow, Brian Collier, Qian Zhang (chair).
 Sep 2023 - Apr 2024 Geometry Seminar, MPI FOR MiS, LEIPZIG, GERMANY
 Co-organizers: Samantha Fairchild, James Farre, Konstantinos Tsouvalas.

CONFERENCES AND WORKSHOPS

Oct 2024 2024 Fall Western Sectional Meeting, conference. UC RIVERSIDE, RIVERSIDE, USA
 Session: *Geometry, topology and dynamics of character varieties*. Co-organizers: Brian Collier.
 Jul 2024 2nd AMS-UMI International Joint Meeting 2024, conference. PALERMO, ITALY
 Session: *Anosov representations and higher Teichmüller theory*. Co-organizers: Sara Maloni, Andrea Tamburelli.
 Aug-Sep 2023 Metrics on higher Teichmüller spaces, workshop. FRÉJUS, FRANCE
 Co-organizers: Christian El Emam, Nathaniel Sagman.
 June 2022 Nearly Carbon Neutral Geometric Topology Conference 2022, online conference. ONLINE
 Session: *Anosov representations*. Co-organizers: Gabriele Viaggi.
 Nov 2021 Virginia Topology Conference, conference. UNIVERSITY OF VIRGINIA
 Co-organizers: Thomas Koberda, Sara Maloni, Mark Pengitore.

PRESENTATIONS

Fall 2021 Reading seminar, UNIVERSITY OF VIRGINIA
An introduction to Higgs bundles, with the Geometry group at UVA.
 Fall 2020-Spring 2021 Reading seminar, UNIVERSITY OF VIRGINIA
Anosov representations, with Sara Maloni and Alec Traaseth.
 Fall 2018 Reading seminar, UNIVERSITÉ DU LUXEMBOURG
A hyperKähler extension of the Teichmüller space, with Andrea Seppi
 Apr 2018 Ph.D. Seminar, UNIVERSITÉ DU LUXEMBOURG
About the McShane identity.
 Sep 2017 Ph.D. Away Days, UNIVERSITÉ DU LUXEMBOURG
Classical applications of the H-Cobordism theorem.
 Sep 2017 Geometry and Topology seminar, UNIVERSITÉ DU LUXEMBOURG
Intertwining operators of the quantum Teichmüller space.
 Apr 2017 Ph.D. Seminar, UNIVERSITÉ DU LUXEMBOURG
A parametrization of complex projective structures on surfaces.
 Sep 2015 Seminari dei Baby Geometri, UNIVERSITÀ DI PISA
Sulle Geometrie in dimensione 3.

SERVICE

Referee for the academic journals *Bulletin of the London Mathematical Society*, *Geometriae Dedicata*, *Geometric and Functional Analysis*, *Mathematische Annalen*, *Advances in Mathematics*, *International Mathematics Research Notices*, *IMRN*, *Journal of Differential Geometry*, *Journal of Topology*, *Algebraic & Geometric Topology*, *Geometry & Topology*, *Transactions of the American Mathematical Society*.

SKILLS

Speaking
Languages

Italian (native), English (fluent), French (intermediate).
C Language, L^AT_EX, Matlab.

REFERENCES

RESEARCH

Francis Bonahon

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Martin Bridgeman

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Sara Maloni

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TEACHING

Thomas Koberda

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